A Refresh-Reduction Digital eDRAM CIM Macro using Asymmetric Error Tolerance Scheme

Yue Cao[1,2,3], Yuanyuan Han[3], Keji Zhou[3], Chengshuo Yu[3], Tianci Cai[1,3], Jianguo Yang[3*]
[1]School of Microelectronics, Frontier Institute of Chip and System, Fudan University, Shanghai, China
[2]Zhejiang Laboratory, Zhejiang, China
[3]Zhangjiang Laboratory, Shanghai, China
Email: yangjg@zjlab.ac.cn (*Corresponding Author)

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***Abstract*—In this work, we propose a refresh-reduction 3T1C eDRAM-based digital Compute-In-Memory (CIM) macro with an asymmetric error tolerance scheme to improve the energy consumption caused by frequent refresh operations. The novel asymmetric error tolerance scheme is first implemented in an eDRAM-based CIM macro to mitigate the effects of the wrong weights, extending the refresh period of the eDRAM cell. 3T1C eDRAM cell exhibits precise asymmetric error behavior, perfectly adapting the proposed asymmetric error tolerance scheme. Additionally, a reconfigurable refresh controller is introduced to eliminate refresh performance overhead during CIM operations and further reduce power consumption by reusing CIM readout data. The proposed 128Kb eDRAM-based CIM macro, simulated at 28nm, achieves a 98-100% improvement in the refresh period and a 21.5% improvement in average energy efficiency at 25°C and 60°C.**

Keywords—eDRAM, DCIM, refresh optimization

I. Introduction

To mitigate the need for frequent off-chip memory access caused by the emerging AI applications, the compute-in-memory (CIM) architecture has been introduced and various types of memory have been explored. Static random-access memory (SRAM) is the most widely used memory in CIM studies due to its fast access speed and high compatibility with advanced CMOS technology [1-6]. However, the limited cell density of SRAM constrains the overall memory capacity. Resistive random-access memory (RRAM) is another promising candidate due to its high density and non-volatile properties [7-11]. Nonetheless, RRAM's inherent resistance variability poses challenges for high-precision computation. Embedded DRAM (eDRAM), on the other hand, offers appealing potential because of its short read/write latency, high cell density, and low power consumption. Several eDRAM-based designs have demonstrated excellent storage density and energy efficiency [12-15].

Despite these advantages, a significant challenge in eDRAM-based CIM design is its limited retention time, which necessitates frequent memory refresh operations [16]. These refresh operations introduce both performance and energy consumption overhead, and this overhead increases with larger array sizes. Previous approaches have sought to address this issue by incorporating metal-oxide-metal (MOM) capacitors at the storage node [15], but the effectiveness of this approach diminishes as fabrication technology scales. Another approach reduces refresh overhead by decreasing the size of individual memory banks and refreshing them in parallel [14], though this comes with additional peripheral circuitry and does not reduce refresh energy consumption. Meanwhile, previous research has indicated that retention errors in eDRAM exhibit an asymmetric pattern, which has been utilized for optimizations in bloom filters or cache refresh [17]. Such a retention error pattern may also be exploited for eDRAM CIM refresh improvement.

To address this challenge, we propose a refresh-reduction digital eDRAM CIM design featuring three key innovations aimed at improving energy efficiency: 1) an asymmetric error tolerance scheme for weight processing, which extends the acceptable refresh period in eDRAM CIM designs; 2) an area-efficient 3T1C eDRAM cell that exhibits precise asymmetric retention error behavior, ensuring the reliability of the proposed tolerance scheme; and 3) a reconfigurable refresh controller that eliminates refresh performance overhead and reduces power consumption by reusing CIM readout data. Our 128Kb design (with 16Kb overhead) is implemented and simulated in 28nm technology.

II. Asymmetric Error Tolerance Scheme

A. Asymmetric Retention Error in eDRAM

As discussed in [17], asymmetry in memory errors refers to the condition where the likelihood of '0-to-1' and '1-to-0' errors differs. In the context of eDRAM design, particularly in conventional 1T1C/2T/3T cell structures, the primary factor influencing this asymmetry is the source-drain leakage current of the write transistor (which introduces single-direction retention errors), while crosstalk effects are negligible during the relatively short refresh periods. As a result, retention errors in these eDRAM cells are highly asymmetric. In standard eDRAM designs, the refresh period is constrained by the worst-case retention time of a few tail bits [18]. By enhancing the asymmetric error tolerance of an eDRAM CIM design, its refresh performance can be significantly optimized. This concept has been previously applied to improve DRAM refresh efficiency [19-20].

B. Asymmetric Error Impact Exploration

First, it is essential to assess the impact of asymmetric errors on the CIM design. Given that '1-to-0' errors are more prevalent in eDRAM structures (such as 1T1C or 3T1C) and that '0-to-1' errors can be processed with a similar method by using inverted data, only '1-to-0' errors will be considered in the following analysis. For the weight-stationary eDRAM CIM design, the asymmetric errors lead to the random resetting of some weight bits to 0. This behavior is somewhat analogous to the dropout layer operation in neural networks, where certain weights are randomly set to 0 during training to reduce overfitting [21]. By applying dropout layers appropriately in convolutional neural networks (CNNs), model generalization can be enhanced without significantly impacting accuracy. Therefore, it is anticipated that CNN model weights should exhibit some tolerance to asymmetric errors without noticeable degradation in performance.

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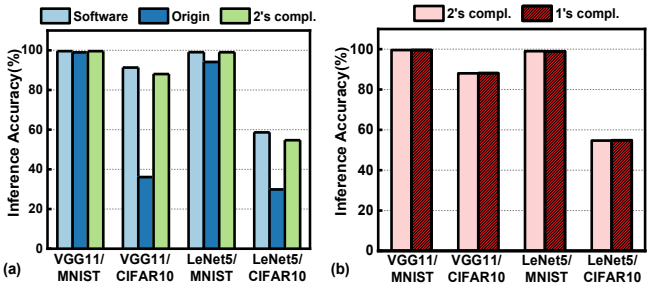


Fig. 1. Inference accuracy evaluation of asymmetric error impact. (a) Comparison between software baseline, origin data, and 2's complement data. (b) Comparison between 2's complement and 1's complement (inverted) data.

Fig. 1(a) illustrates the inference accuracy of the software baseline compared to various conditions affected by asymmetric error. To evaluate the impact of asymmetric error, we selected two CNN INT8 models, VGG11 and LeNet5, along with two datasets, CIFAR10 and MNIST, applying an asymmetric error probability of 5%. Contrary to initial expectations, the results show that directly applying asymmetric error to the original data leads to a significant accuracy drop in most cases. A potential reason for this is that when signed weights are affected by asymmetric errors, the impact is not limited to attenuation as with dropout layers. Instead, the error can cause inversion (for the sign bit), attenuation (for positive values), or amplification (for negative values), resulting in a much higher level of randomness in the effect.

Building on this observation, the key to improving weight error tolerance is ensuring that asymmetric errors only result in attenuation effects. A straightforward solution is to store weights in absolute value form and then recover them to their signed version during computation. Fig. 1(a) also includes accuracy data for this absolute value approach, based on the 2's complement format. The results demonstrate a significant restoration of inference accuracy, confirming that this weight storage method provides better tolerance to asymmetric errors.

C. *Proposed Asymmetric Error Tolerance Scheme*

Building on the evaluation above, we can further optimize the weight processing scheme to simplify the design. First, recovering data in 2's complement format introduces an additional addition during data readout, which incurs considerable area and latency overhead. To eliminate this overhead, we propose storing weights using 1's complement (inversion) data instead. This approach is supported by the simulation results shown in Fig. 1(b), which indicate that the complement processing method has minimal impact on final inference accuracy. As inverted data can be directly generated by the differential sense amplifier, the associated overhead is significantly reduced.

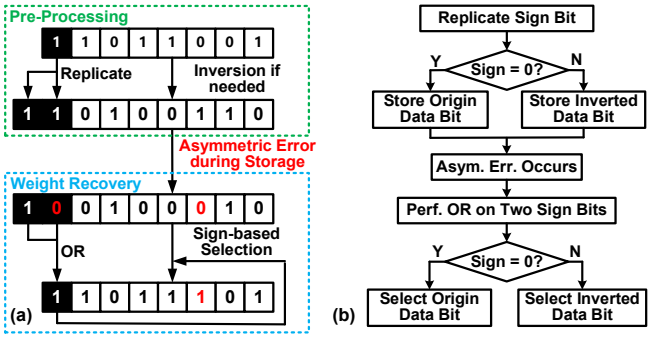


Fig. 2. Proposed asymmetric error tolerance scheme (a) example and (b) workflow for weight pre-processing and recovery.

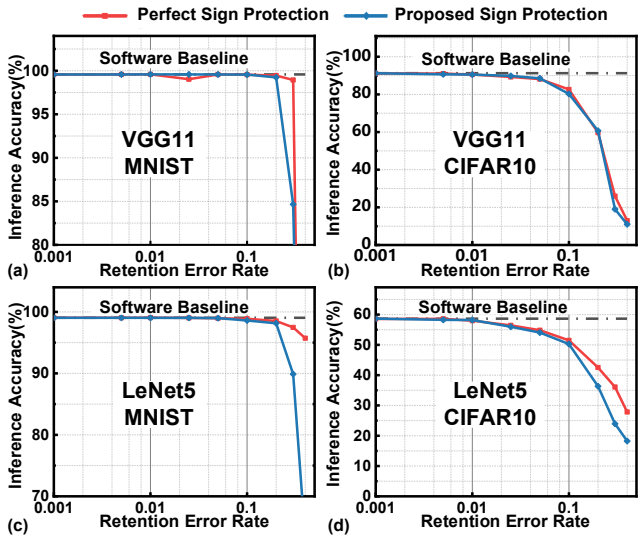


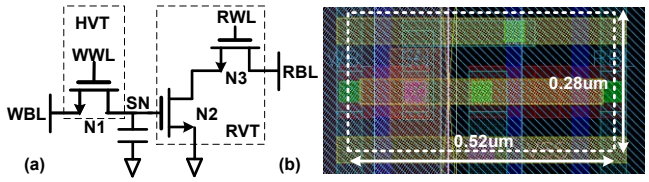
Fig. 3. Asymmetric error impact comparison between perfect sign protection and proposed sign protection method across different retention error rates. (a) VGG11/MNIST (b) VGG11/CIFAR10 (c) LeNet5/MNIST (d) LeNet5/CIFAR10.

Second, previous simulation results emphasize the importance of sign bit accuracy for retention error tolerance. To ensure this, a correction or protection scheme for the sign bit must be considered. Error correction code (ECC) is widely used for memory data correction due to its high reliability, but it is accompanied by complex implementation costs and significant area and latency overhead [22-23]. Since CIM designs require fast data decoding, we propose a simpler sign protection scheme using sign bit replication.

Fig. 2 illustrates the process and workflow of our proposed asymmetric error tolerance scheme. During the pre-processing stage, the sign bit is first replicated into two bits. For the remaining data bits, if the sign bit is 1, the bits are inverted; if the sign bit is 0, the bits remain unchanged. The pre-processed weight data is then stored in the eDRAM. When the data is read out, the sign bit is recovered using an OR operation on the two replicated sign bits. Then, the rest of the data bits are determined: if the sign bit is 1, the inverted version of the read data is selected; otherwise, the original version is used. For INT8 weights, the area overhead of this method is approximately 12.5%, which can be further reduced as data precision increases. Assuming the probability of a retention error is p, the probability of a sign bit error is approximately $0.5p^2$ (assuming 50% of weights are negative). When p is small, the probability of a sign bit error becomes negligible.

The effectiveness of the proposed scheme is demonstrated in Fig. 3, which compares the inference accuracy of perfect sign protection versus the proposed sign protection scheme across different retention error rates. The results show that when p is small, the accuracy drop for both conditions is similar. As p increases and leads to divergence, the accuracy loss becomes too significant for practical use. Therefore, the proposed sign protection scheme is adequate for our asymmetric error tolerance scenarios. Given that the baseline accuracy for LeNet5/CIFAR10 is too low for practical applications, and VGG11/MNIST might be over-engineered, we focus on using LeNet5/MNIST and VGG11/CIFAR10 for subsequent design evaluations, which

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Mode	WWL	WBL	RWL	RBL
Idle	-0.1V	0V	0V	0.9V
Write 0	0.9V	0V	0V	0.9V
Write 1	0.9V	0.9V	0V	0.9V
Read	-0.1V	0V	0.9V	Float

(c)

Fig. 4 Proposed 3T1C eDRAM cell. (a) Schematic of the 3T1C cell. (b) Layout of the 3T1C cell with size of 0.1456μm^2. (c) Cell operation table.

also gives similar ideal inference accuracy with the state-of-art designs [12-15] for better comparison.

III. HARDWARE IMPLEMENTATION

A. 3T1C eDRAM Cell with Asymmetric Retention Error

Fig. 4 presents the proposed 3T1C eDRAM cell and its operation principles. As shown in Fig. 4(a), the cell consists of a high-threshold voltage (HVT) NMOS write transistor and two regular-threshold voltage (RVT) read transistors, along with an additional metal-oxide-metal (MOM) capacitor for improved retention. All transistors are minimum-sized for higher cell density. Fig. 4(b) depicts the layout of the cell, which has a size of 0.1456 μm², based on 28nm technology. Fig. 4(c) provides the operation table for this cell. During idle or read operations, a negative voltage of -0.1V is applied to the write word line (WWL) to suppress leakage current, and the read bit line (RBL) is precharged to 0.9V during idle states to accelerate read operations.

The retention time of the proposed 3T1C eDRAM cell was evaluated at 25°C and 60°C, with the results of a 6K Monte Carlo simulation shown in Fig. 5(a) and (b). The results indicate that the storage node (SN) voltage gradually decreases, crossing the 380mV threshold for the "1" state, while the SN voltage for the "0" state remains below 0.1V throughout the evaluation period. This behavior aligns with the asymmetric error assumption that underpins the proposed weight processing scheme. Fig. 5(c) illustrates the relationship between single-bit error rates and retention time. The results reveal that, for both 25°C and 60°C conditions, tail bits exhibit significantly shorter retention times compared to the majority of memory cells. Thus, enhancing retention error tolerance can substantially extend the

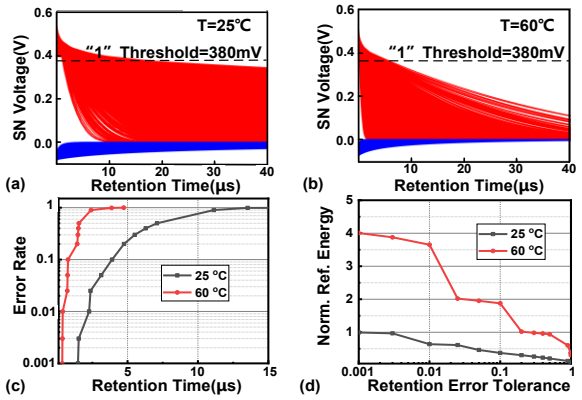


Fig. 5 Retention analysis of the proposed cell. 6K Monte Carlo simulation for retention time at (a) 25℃ and (b) 60℃. (c) Single-bit error rate at different retention times. (d) Normalized refresh energy overhead at different error tolerance rates.

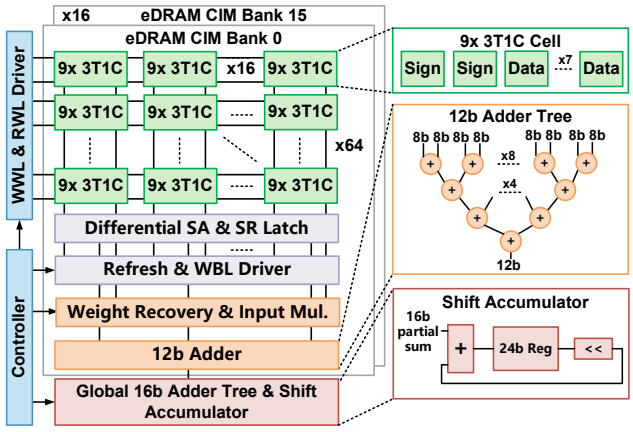


Fig. 6 Architecture overview of the proposed eDRAM CIM design.

acceptable refresh period and reduce refresh energy overhead, as shown in Fig 5(d).

B. Design Architecture Overview

The proposed digital eDRAM CIM architecture is shown in Fig. 6. It consists of 16 eDRAM CIM banks, each with a 144×64 eDRAM subarray, where every 9 cells (two sign bits and 7 data bits) in a row store one weight unit. Each bank includes a differential sense amplifier (SA) & SR latch stage, a refresh and write bitline (WBL) driver stage, a weight recovery and input multiplier stage, and a 12-bit adder tree. Details of the CIM datapath with weight recovery are discussed in Section III.C. The 12-bit partial sums generated by the CIM banks are sent to the global 16-bit adder tree for final summation. Input activation is transferred in bit-serial form (MSB-to-LSB), and the partial sums are accumulated in a shift register to produce the final result.

C. CIM Computation Datapath with Weight Recovery

Fig. 7 shows the schematic and timing waveform of the CIM computation datapath with weight recovery. For simplicity, only two sign bit cells (8th and 7th bits) and one MSB data bit (6th bit) for a single weight storage are depicted. The weight data is pre-processed and stored according to the scheme outlined in Section II.

Before the read operation begins, PRE_N is set to low, and all read bit lines (RBLs) are precharged to 1. When a row of eDRAM cells is selected, the corresponding RWL is set to high. If a cell stores a "1," the associated RBL discharges. Once sufficient voltage difference is established, SAEN is set to high, and the differential SA compares the

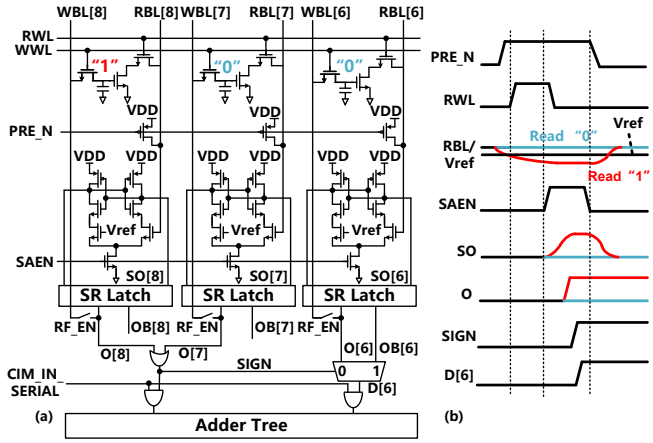


Fig. 7 CIM computation datapath overview. (a) Schematic of the CIM datapath. (b) Timing waveform of the related signals.

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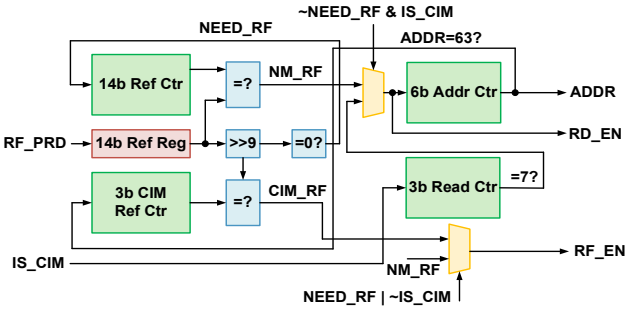


Fig. 8 Block diagram of the reconfigurable refresh controller.

RBL voltage with the reference voltage (Vref). If RBL is less than Vref (indicating a "1" read), SO is set to 1. Otherwise, SO is set to 0. RWL is then disabled to reduce RBL discharge energy. The SA outputs are transferred to the SR latch, and if a refresh is needed, RF_EN is enabled for data refresh. Once the readout data is stored in SR latches, the sign value is generated by an OR operation between O[8] and O[7], with 2-to-1 multiplexers selecting the data bits. If the sign is 0, the original data is chosen (like O[6]); otherwise, the inverted data is selected (like OB[6]). Finally, the generated data is multiplied by the bit-serial input using an AND operation before being sent to the adder tree.

D. Reconfigurable Refresh Controller

To enable efficient refresh operations during CIM calculations, we propose a reconfigurable refresh controller, as shown in the block diagram in Fig. 8. Since CIM operations involve reading every row in the memory subarray, if the refresh period is larger than one CIM iteration time, we can reuse the CIM readout data for refreshing during execution, thereby eliminating the performance overhead associated with refresh operations. Additionally, to reduce critical timing delays, we leverage the dual-port capability of the 3T1C cell to perform refresh writeback in the cycle following the read operation.

In the proposed controller, the refresh period is reconfigurable by writing the desired period (measured in clock cycles) into a 14-bit refresh register. By pipelining read and compute operations, each row computation takes 8 cycles, and a full CIM iteration then requires 512 cycles. Therefore, to achieve comparison, the controller uses a 9-bit right shifter to divide the refresh period by the CIM iteration time. If the output is 0, or if CIM is not executing, the 14-bit refresh counter controls the refresh operation. When the counter reaches the specified refresh cycle number, CIM operations are suspended, and a normal refresh begins. Once the refresh is complete, the CIM flow resumes. If the output is equal to or larger than 1, in-CIM refresh can be executed.

During CIM execution, the controller initiates a read operation and increments the address every 8 cycles using a 3-bit read counter. When the address counter reaches the last row (row 63), the 3-bit CIM refresh counter increments. Once the refresh counter matches the 9-bit right shifter output, a refresh is enabled after a read operation. This design allows the controller to manage refresh during CIM execution with a reconfigurable period, avoiding performance overhead from additional refresh cycles.

IV. SIMULATION RESULTS

The simulation results of our eDRAM CIM design are shown in Fig. 9. Fig. 9 (a) and (b) demonstrate the relationship between the inference accuracy and macro refresh period for the origin baseline and our proposed design at 25°C and 60°C. Allowing for a 3% accuracy drop, the maximum refresh period was improved by over 98–100% under both temperature conditions, which gives a 21.5% energy efficiency improvement on average. The refresh period was extended to 3.13–4.75µs at 25°C and 0.72–1.42µs at 60°C. This improvement yields an energy efficiency of 10.22–10.71 TOPS/W at 25°C. Average power breakdown is presented in Fig. 9 (d). Table I compares this design with state-of-the-art implementations, demonstrating that it achieves the long refresh period for an area-efficient cell structure while eliminating refresh performance overhead. These results confirm the success of our refresh optimization in the proposed CIM macro.

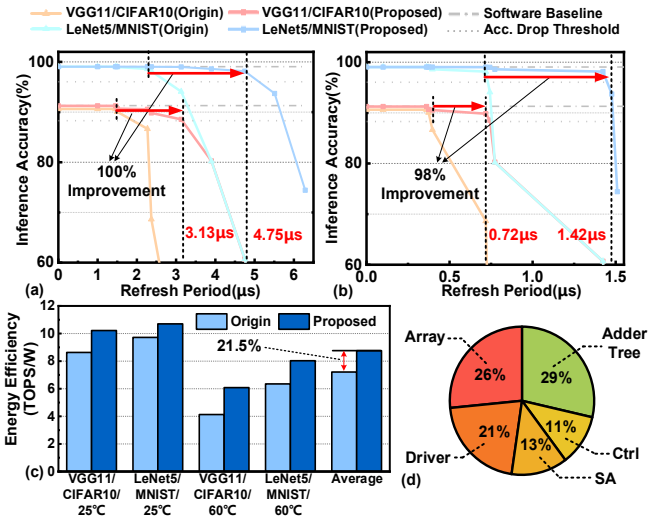


Fig. 9 Simulation results of the proposed design. Inference accuracy comparison between the original hardware baseline and the proposed macro at (a) 25℃ and (b) 60℃. (c) Energy efficiency comparison. (d) Average power breakdown.

V. CONCLUSION

This paper presents a refresh-reduction digital eDRAM CIM design, simulated using 28nm technology. The simulation results demonstrate that the proposed asymmetric error tolerance scheme significantly extends the acceptable refresh period across different temperatures and CNN applications. This scheme is expected to be applicable to other eDRAM cell structures exhibiting asymmetric retention error behavior. Additionally, the reconfigurable controller minimizes the refresh overhead by reusing CIM readout data and performing refresh writeback during computation. By reducing eDRAM refresh operations, this design enhances system performance and energy efficiency, showcasing the potential for large-scale CIM designs based on eDRAM cells.

Table I. DESIGN COMPARISON

	ISSCC'21[12]	ISSCC'24[13]	VLSI'22[14]	TCAS'21[15]	This work
Technology	65nm	28nm	65nm	65nm	28nm
Bit Cell	1T1C eDRAM	3T eDRAM	2T1C eDRAM	4T2C eDRAM	3T1C eDRAM
Cell Area(µm²)	22.08	0.145	N/A	1.08	0.146
Refresh Period	> 2ms	1.3µs	N/A	200µs	3.13~4.75µs
Array Size	16Kb	40Kb	32Kb	16Kb	128Kb
Input/Weight Bits	8/8	8/8	8/8	3/3	8/8
Ref. Perf. Overhead?	N/A	Yes	N/A	Yes	No
Energy Efficiency (TOPS/W)	4.71	18.1	7.39	17.8	10.22~10.71
Model/Dataset	ResNet-20 /CIFAR10	N/A	ResNet-20 /CIFAR10	MLP/MNIST CNN/CIFAR10	LeNet5/MNIST VGG11/CIFAR10
Inf. Accuracy	90.77%	N/A	92.2%	96.78%/82.8%	98.15%/88.55%
FoM[a]	304.6	1158.4	473	160.2	655~685

a. FoM = Energy Efficiency * Input Bits * Output Bits

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